

Optimizing EEG Signal Classification for Motor Imagery BCIs: FilterBank CSP with Riemannian Manifolds and Ensemble Learning Models

M. Moein Esfahani

Department of Mechanical Engineering
K. N. Toosi University of Technology
Tehran, Iran
esfahani.moein@email.kntu.ac.ir
esfahani.moein@gmail.com

M. Hosein Najafi

Department of Computer Engineering
Islamic Azad University
Tehran, Iran
najafi.m.hosein@gmail.com

Hossein Sadati

Department of Mechanical Engineering
K. N. Toosi University of Technology
Tehran, Iran
Sadati@kntu.ac.ir

Abstract— Motor imagery (MI) is a compelling cognitive phenomenon situated at the intersection of cognitive neuroscience and technology, offering insights into human cognition and a spectrum of practical applications. Electroencephalography (EEG) emerges as a technology, offering a non-invasive means to record electrical activity within the brain. Motor imagery-based Brain-Computer Interfaces (MI-BCIs) enables the translation of cerebral activity into implementable commands found in the sensorimotor areas of the brain. This paper offers a comprehensive exploration of MI, delving into its practical implications, the complexities it entails, and the critical role this research focuses on enhancing the classification accuracy of EEG signals with Motor Imagery task. We employ a comprehensive approach that integrates Filter Bank Common Spatial Pattern (CSP) and the Riemannian manifold to extract features, aiming at maintaining critical signal details. Furthermore, we utilize Linear Discriminant Analysis (LDA) alongside advanced classification methods, such as AdaBoost and Total Boost, to elevate classification accuracy. Our results, reveal significant improvements in the classification accuracy. This research contributes to the advancement of MI EEG signal classification and opens new avenues for applications in fields such as brain-computer interfaces and medical diagnostics.

Keywords— BCI, EEG, Motor-Imagery, CSP, Filter-Bank, Ensemble learning, Ensemble Learning, LDA.

I. INTRODUCTION

The realm of Brain-Computer Interfaces (BCIs) stands at the forefront of contemporary technology, offering an innovative pathway for users to communicate with computers through the exclusive use of brain activity, predominantly monitored via Electroencephalography (EEG) [1], [2]. This technological concept extends a wide array of applications, promising to revolutionize various aspects of human-computer interaction[1]–[3][4]. From providing a lifeline for individuals with severe physical disabilities to serving as innovative control mechanisms for gaming and fostering adaptive human-computer interfaces responsive to users' mental states, BCIs present endless possibilities. However, despite the tremendous promise, BCIs have not yet achieved widespread integration beyond the laboratory setting[5], [6]. A central barrier to the broader application of EEG-based BCIs is their vulnerability to low robustness and reliability, leading to frequent misinterpretation of user commands and reduced accuracy, even in controlled laboratory conditions[6], [7]. One of the primary challenges that BCIs face is the considerable time and effort required for calibration and

training. This necessity arises from the need to use a substantial dataset of training EEG samples from each user and fine-tune the BCI system to maximize performance for each individual[6], [7][8][1]–[3], [9]. By recording MI data non-invasively through EEGs and converting them into implementable commands, these BCIs could significantly improve the quality of life and mobility of individuals affected by physical disabilities. To appreciate the significance of BCIs and EEGs in the domain of motor imagery, it is essential to explore the significance of improving human brain signals. This technology not only holds the potential for diagnosing various diseases, but also serves as a valuable aid in the performance of physical tasks, from rehabilitation to recreational activities and device control. By integrating BCIs and EEGs, it becomes possible to record, analyze, and transform brain signals into executable commands associated with an output device[1]–[3], [9]–[11]. The human brain comprises two hemispheres, each responsible for a distinct set of cognitive and motor functions. This division is pivotal in designing BCIs that align with the user's motor intentions, offering more intuitive and efficient control mechanisms. Furthermore, this exploration delves into the role of central beta and mu rhythms in motor imagery tasks[1], [2][12]. These rhythms provide valuable insights into how they can be harnessed to explore an array of voluntary movements throughout the body, including movements of the fingers, tongue, and feet. This strategic focus on feature extraction and signal processing techniques for the classification of motor imagery tasks, unveils crucial insights into effective features that can significantly enhance the functionality and rehabilitative potential of EEG-based BCIs. Through this paper, our goal is to contribute to the broader understanding of EEG BCIs in the context of motor imagery tasks, offering insights that may facilitate technological advancements and enhance the quality of life for those in need[13].

In this study, our primary objective is to introduce our novel methodologies, offering insights into our approach to EEG signal processing and our strategies for enhancing classification performance. We provide a comprehensive discussion of the Filter Bank Common Spatial Pattern (CSP) technique and the application of Riemannian geometry, highlighting their application within the scope of our research. Additionally, we delve into the concept of EEG signal processing, their roles, and importance within our analytical framework. Moreover, we introduced the dataset that forms the basis of our study. Ultimately, our intention as authors in the discussion section is to explain provide our readers with an understanding of the implications, significance, and potential

avenues for future research that stem from our diligent efforts in this study.

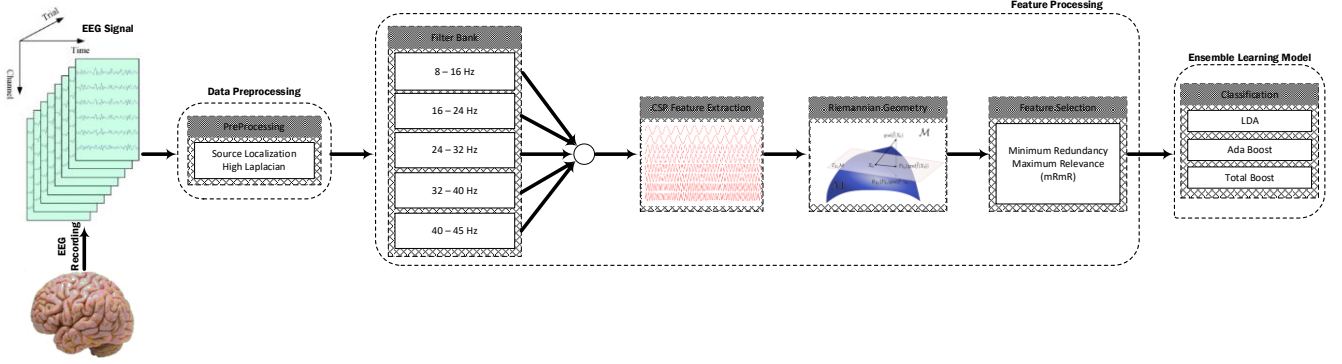


Fig. 1. The schematic of the EEG signal classification approach during motor imagery tasks

II. METHODOLOGY

In the context of this research, we present our cutting-edge methodology, as visually depicted in Figure 1. The principal aim of our research is the classification of EEG signals with motor imagery tasks utilizing innovative techniques. Historically, numerous studies and methodologies have been put forth to address this task [12]–[14][1]–[3], [9]. Nevertheless, with our novel approach and method, we aspire to yield superior results and enhance the classification performance.

A. EEG Signal with Motor Imagery Task

Motor Imagery (MI) as recorded and decoded from human brain signals stands as an important non-invasive cognitive task, holding the promise of multivarious diagnostic and operational applications, such as disease diagnosis and facilitation of various physical tasks encompassing rehabilitation, recreation, device manipulation, and technological assistance. Within this context, Brain-Computer Interfaces (BCIs) emerge as the main system for recording brain signals through non-invasive Electroencephalograms (EEGs), followed by the intricate analysis and subsequent translation of these neural manifestations into implementable commands related to the intended output device. The human brain, characterized by its intricate divisions into two hemispheres, the right and the left, exhibits a notable feature in terms of control and functionality. Notably, the left hemisphere governs motor control over the right side of the body and is intricately linked with duties encompassing creativity, spatial orientation, imagination, emotional processing, and multitasking[7], [12], [14], [15]. In contrast, the right hemisphere performs over the left side of the body, primarily engaged in logical reasoning and the realms of scientific and mathematical thinking[13], [16]. Moreover, the brain can be anatomically delineated into four cardinal lobes, namely the frontal, parietal, temporal, and occipital lobes. The frontal lobes are instrumental in governing behavior, executive functioning, and problem-solving tasks, while the parietal lobes are primarily responsible for processing sensory inputs and interpreting the perceptual reality. The occipital lobe is specialized in the reception and initial processing of visual stimuli, which are subsequently relayed to other cerebral regions. On the other hand, the temporal lobe assumes a critical role in memory

consolidation and serves as the epicenter for auditory perception and language processing. This complex neural landscape forms the substrate for the intricate orchestration of human motor imagery[7]. These neural oscillations facilitate the exploration of imagined voluntary movements in various anatomical regions, encompassing areas such as fingers, tongue, and feet. The outcomes of motor imagery delineate a systematic engagement of discrete segments within the primary motor cortex, facilitating specific neural representations of distinct non-motor regions. However, the pursuit of effective feature extraction and signal processing techniques remains a paramount concern.

B. Filter Bank - Common Spatial Pattern (FBCSP)

The Filter Bank Common Spatial Pattern (FBCSP) model approach, as illustrated in Figure 1, represents a sophisticated method for the analysis of electroencephalography (EEG) signals with Motor Imagery task[17]. This proposed approach is characterized by a series of four progressive stages aimed at enhancing the processing of EEG data, each of which plays a crucial role in extracting valuable information for a wide range of applications, particularly in the field of Brain-Computer Interfaces (BCIs).

In the initial stage of the FBCSP process, a filter bank is applied to the EEG measurements. This filter bank employs multiple bandpass filters based on the zero-phase Chebyshev Type II Infinite Impulse Response (IIR) filter design. Zero-phase filtering is instrumental in ensuring the preservation of signal integrity by compensating for the nonlinear phase shifts induced by IIR filters. The application of these bandpass filters results in the segmentation of EEG data into distinct frequency bands, thereby enabling the isolation of relevant neural activity across the spectrum.

The second stage of the FBCSP approach introduces spatial filtering through the Common Spatial Pattern (CSP) algorithm. This step involves applying the CSP algorithm to each of the segmented frequency bands obtained in the first stage. The CSP algorithm effectively isolates spatial patterns and spatial filters that are highly discriminative for the specific frequency range of the respective bandpass filter. This not only enhances the specificity of the spatial filters but also refines the extraction of neural features from the EEG data.

The third stage of the FBCSP method involves feature selection. Here, a feature selection algorithm is employed to discern the most discriminative CSP features from the filter bank. This selection process is critical in identifying the neural patterns and characteristics that are most relevant to the task at hand, whether it be for BCI control, cognitive state classification, or any other EEG-based application. The discriminating features thus identified are essential for enhancing the accuracy and efficiency of subsequent data analysis and classification tasks.

C. Common Spatial Pattern(CSP)

The first introduction of the Common Spatial Pattern (CSP) algorithm goes back to the process of classification of discriminative spatial patterns from multichannel EEG signals specifically with motor imagery tasks[17][1], [2]. CSP's base principle resides in its capacity to turn the EEG channels into a new spatial space where it maximizes the variance associated with one class while minimizing that of another. Consequently, this yields spatial filters that effectively enhance neural activity patterns corresponding to specific cognitive states while suppressing unrelated activities. The benefits of the CSP algorithm are many, including the reduction of redundant or noisy data. Using this method, two distinct spatial filters can be derived for the extraction of task-related signal components while concurrently eliminating both task-independent components and noise. In this study, to optimize the CSP method, we employed a state-of-the-art method to use CSP and Riemannian manifold to achieve better results. It is important to know that CSP performs as part of the feature extraction method in this study. Also, it is noteworthy that the algorithm is basically designed for datasets with only two distinct classes. CSP uses spatial filters that extremize the following cost function:

$$J(w) = \frac{w^T C_1 w}{w^T C_2 w} = \frac{w^T X_1 X_1^T w}{w^T X_2 X_2^T w} \quad (1)$$

Where the superscript 'T' represents the transposition operation, which transforms the signal trial for class I into a matrix format. In this format, the rows correspond to channels, while the columns correspond to discrete time points. At the core of our approach lies Eq (1), an essential component of our spatial covariance matrix for class I, while our objective is to maximize this cost function.

Assume the two-class motor imagery signals denoted as $X_{11}, \dots, X_{1n}, X_{21}, \dots, X_{2m}$ where the subscripts 'n' and 'm' respectively denote the number of trials for each class. The covariance matrices are

$$C_{1i} = \frac{1}{\text{trace}(X_{1i} X_{1i}^T)} X_{1i} X_{1i}^T \quad \text{where } i = 1, \dots, n \quad (2)$$

$$C_{2j} = \frac{1}{\text{trace}(X_{2j} X_{2j}^T)} X_{2j} X_{2j}^T \quad \text{where } j = 1, \dots, m \quad (3)$$

It is important to consider that the Common Spatial Pattern (CSP) methods, despite their extensive use and demonstrated effectiveness, are not without their limitations. One of the primary drawbacks is their

heightened sensitivity to noise, a factor that can potentially lead to overfitting, especially when dealing with datasets containing a limited number of trials. To address these limitations, we proposed using Riemannian manifold approach with CSP method as a novel method.

D. Riemannian Manifold

The Riemannian manifold, defined within the space of positive definite and symmetric matrices, constitutes a special geometric framework[7], [12], [15]. This manifold, characterized by the Riemannian distance, is both differentiable and non-Euclidean in nature. Particularly, when dealing with positive definite and symmetric matrices, the Riemannian distance offers a more precise means of measuring the interrelationships between these matrices in comparison to the Euclidean distance. Notably, in cases where two matrices have significant similarity, differentiating between them using either the Riemannian distance or the Euclidean distance becomes a challenging task. However, it is essential to note that in scenarios where the matrices exhibit only slight dissimilarity, the classification results achieved through the Riemannian distance tend to outperform those attained via the Euclidean distance.

Pseudocode for EEG Signal Classification using Riemannian Manifolds:

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Import EEG Data (segmented)
Import EEG Labels

1. Split_Data(EEG data, Labels):
2.   Evaluate estimate covariances with CSP
3.   Calculate Riemannian Manifold (cov matrices_train)
4.   Calculate initialize_classifier()
5.   Calculate Classification Performance
6. End

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Fig. 2. Pseudo code of the proposed model

$$d_R(A, B) = \left\| \log(A^{-\frac{1}{2}} B A^{-\frac{1}{2}}) \right\| \quad (4)$$

, where you can calculate Riemannian distance.

$$P_{med}^{(k+1)} = R_e(P_{med}^{(k)}, V^{(k)}) \quad (5)$$

where it is Riemannian geometric median. And based on the pseudo code in Figure 2 we use feature extraction function with Riemannian manifold as

$$S_i = \text{upper}(\log(P_G^{-\frac{1}{2}} P_i P_G^{-\frac{1}{2}})) \quad (6)$$

where P_i represents the covariance matrix. The results obtained from Eq. 6 are features that can be used in the selection of optimal feature set.

E. Feature selection with mRmR

The Minimum Redundancy Maximum Relevance (mRmR) method is a technique for feature selection[18]–[20]. It operates by evaluating the relevance of each feature to the target variable while also considering the redundancy among features. In essence, mRMR aims to identify a subset of features that are highly informative for predictive

modeling while minimizing the inclusion of redundant features. By optimizing a balance between maximizing the relevance of selected features and minimizing their redundancy, the mRmR method can significantly enhance the efficiency and accuracy of ML models. This approach has found applications in various domains and former studies, where feature selection is crucial for improving model performance, reducing computational costs, and enhancing the interpretability of results.

F. H-Laplacian Source Localization

The main objective of this research focusses on the analysis of EEG signals with motor imagery task. Derived from the authors' prior research, this study incorporates the High Laplacian spatial filter as a method for source localization[1], [2]. By employing this approach, our aim is to improve and optimize the signal-to-noise ratio (SNR) with the ultimate goal of enhancing the classification performance of EEG signals. In contrast to other publications by the same authors[1]–[3], [9], where the application of common average reference (CAR) alongside with either low or large Laplacian spatial filters leading to a reduction in SNR, our present study tend to achieve the opposite effect. This method was accomplished by calculating the weighted sum of potentials recorded by adjacent electrodes, and subsequently subtracting this sum from the potential measured at the electrode of interest.

TABLE I. PERFORMANCE COMPARISON FOR OTHER STUDIES.

Subject Num	Feet Label ref	Feet Label	Tongue Label ref	Tongue Label	Average ref	Average
Subject 1	91.66	90.42	83.33	90.42	87.5	90.42
Subject 2	83.33	75.55	95	85.33	89.17	80.44
Subject 3	100	92.31	83.33	82.41	91.67	87.36
Subject 4	91.66	95	100	100	95.83	97.5
Subject 5	91.66	95	83.33	85.42	87.5	90.21
Subject 1	97.77	96.33	88.88	95.32	93.33	95.825
Subject 2	66.66	82.41	83.33	85.46	75	83.935
Subject 3	95.83	92.22	95.83	97.32	95.83	94.77

G. Classification

The efficacy of the feature vector extracted from EEG trials plays an important role in classification. One of the challenges in this area is around finding the optimal set of observation classes based on the training dataset, wherein class labels are readily available. In recent studies focusing on Motor Imagery (MI) classification and Linear Discriminant Analysis (LDA) has employed as a statistical classification technique, particularly for the analysis of EEG data. Its main purpose lies in building a predictive model that performs with hyperplanes. These hyperplanes are important in creating a linear discriminant function that maximizes the ratio of between-class variance to within-class variance, thereby enhancing the separation of data points. Based on Eq. 7, LDA efficiently addresses the challenge of classifying two-tier training data, taking into account the variance inherent in the dataset.

$$j(w) = \frac{(m_1 - m_2)^2}{(s_1^2 + s_2^2)} \quad (7)$$

AdaBoost, stands as an ensemble machine learning algorithm widely used to an array of classification and regression tasks. Within the framework of supervised learning[9], AdaBoost is employed in classification by combining several weak or base learners, such as decision trees and LDA, into an advanced combined learner[1], [9]. The essence of AdaBoost lies in the dynamic weighting of instances within the training dataset, with the weights adjusted based on the accuracy of previous classifications. The invention of AdaBoost as an ensemble modeling approach traces back to its conceptualization by Freund and Schapire in 1997[1], [9][21]. Over the years, boosting has gained immense traction as an effective strategy for addressing binary classification challenges. These boosting algorithms optimizes the potential of ensembling by elevating the predictive power of multiple weak learners to the status of strong learners. The main operational principle of boosting algorithms revolves around an iterative process. Initially, a model is constructed on the training dataset, and subsequently, a second model is built to refine the faults and limitations of the initial model. This iterative process continues until the errors are minimized, and the dataset can be accurately predicted. It is important to note that boosting algorithms operate by merging multiple models, the so-called weak learners, to yield the final output represented by the strong learners. This iterative approach within boosting algorithms results in the creation of a model and the equal value of weights to all data points. Subsequently, greater weights are assigned to data points that have been with higher classification results. Figure 3 accounts for these weighted instances, giving priority to those with higher weights. This iterative training process continues until a smaller error is achieved, signifying an enhanced predictive accuracy.

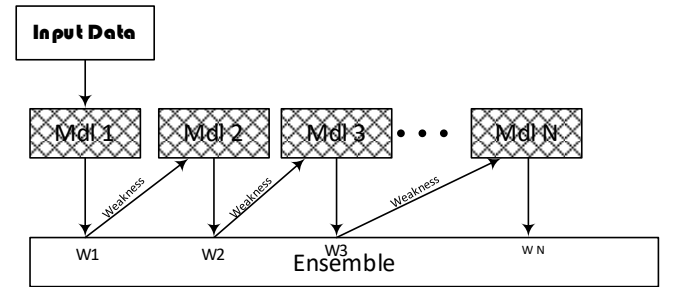


Fig. 3. Proposed ensemble learning model

TABLE II. PERFORMANCE COMPARISON FOR MULTIPLE CLASSIFICATION MODELS

Subject Num	Average LDA	Average AdaBoost	Average Total Boost
Subject 1	90.42	95.48	83.42
Subject 2	80.44	91.23	79.45
Subject 3	87.36	89.46	88.36
Subject 4	97.5	97.5	89.32
Subject 5	90.21	93.25	88.35
Subject 1	95.825	94.38	92.42
Subject 2	83.935	85	78.32

Subject 3	94.77	95.42	89.32
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III. DATASET

In our study, we conducted the evaluation and the performance of our proposed method using publicly available BCI-EEG datasets, specifically the BCI Competition IV benchmark dataset IIa (BCI IV-IIa) and the BCI Competition III benchmark dataset IIIa (BCI III-IIIa)[22]. These datasets are recorded using motor imagery EEG (MI-EEG) signals associated with a variety of task movements. It includes those involving the left and right hands, both feet, and the tongue. The data were collected from 60 EEG channels, with each class including 60 individual trials. The acquired EEG signals were discretized at a rate of 250 Hz and filtered within the frequency range of 1 Hz to 50 Hz. The data collection process involved three participants engaging in motor imagery tasks.

IV. RESULTS AND DISCUSSION

The primary objective of this study is to optimize the classification performance of Motor Imagery (MI) EEG signals. As previously mentioned, prior research utilized the Common Spatial Pattern (CSP) and its optimized variations. In this research, we have employed the Filter Bank CSP technique to extract additional features and avoid losing signal details. Additionally, we integrated the Riemannian manifold approach to extract features from covariance matrices derived from the CSP method. Subsequently, we applied the Linear Discriminant Analysis (LDA) classification method along with AdaBoost and Total Boost as advanced techniques to further enhance classification performance. The results, as presented in Table 1, indicate notable improvements for specific subjects within both datasets when employing the LDA model. Table 1 and 2 are divided into two primary sections: the initial five rows pertain to the first dataset, while the subsequent three rows correspond to the second dataset. The overall and average accuracy metrics exhibit enhancements with the application of the LDA model. Furthermore, we compared our results with those from previously published studies, revealing that our approach outperforms the prior research. Table 2, on the other hand, offers a comparative analysis of the three classification models used in our study. LDA, identical to its presentation in Table 1, is joined by AdaBoost and Total Boost models. The results indicate that Total Boost yields the lowest classification accuracy, with AdaBoost slightly outperforming LDA. It is noteworthy that the accuracy measures in our study are obtained through a K-fold cross-validation with K=10, a strategy employed to mitigate overfitting and ensure the accuracy of our results.

V. CONCLUSION

This research endeavor has focused on the optimization of Motor Imagery (MI) EEG signal classification, with integration of Filter Bank CSP and Riemannian manifold techniques. Our goal was to enhance classification accuracy and to ensure that we do not lose signal details. By utilizing Linear Discriminant Analysis (LDA) alongside advanced classification methods, such as AdaBoost and Total Boost, we have strived to achieve superior classification results. This research not only contributes to the evolving field of EEG signal classification, but also highlights the potential for enhanced accuracy and performance through the

integration of novel techniques. The findings presented here offer promising results for future studies in the research field of MI EEG signal analysis, with potential applications ranging from brain-computer interfaces to medical diagnostics and beyond.

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